INCH–POUND

MIL–C–87264 (USAF)
11 February 1992

MILITARY SPECIFICATION

COVERALLS, CHEMICAL DEFENSE CWU–66/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fire–resistant, aircrew, chemical protective coveralls.

1.2 Classification

1.2.1 Type. The coveralls shall be furnished as Type CWU–66/P — sage green as specified (see 6.2).

1.2.2 Sizes. The coveralls shall be furnished in the following sizes, as specified (see 6.2):

Schedule of Sizes

32S	36S	38L	42R	46S
32R	36R	40S	42L	46R
34S	36L	40R	44S	46L
34R	38S	40L	44R	48R
	38R	42S	44L	48L

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright–Patterson AFB OH 45433–6503 by using the self–addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN–P–71	–	Pallet, Material Handling, Wood, Stringer Construction, 2–Way and 4–Way (Partial)
V–F–106	–	Fastener, Slide, Interlocking
DDD–L–20	–	Label: For Clothing, Equipage, and Tentage (General Use)

MILITARY

MIL–B–117	–	Bags, Sleeves and Tubing
MIL–B–131	–	Barrier Materials, Watervaporproof, Greaseproof, Flexible, Heat–Sealable
MIL–W–4088	–	Webbing, Textile, Woven Nylon
MIL–C–5040	–	Cord, Nylon
MIL–W–5664	–	Webbing, Textile, Elastic
MIL–P–15011	–	Pallets, Material Handling, Wood Post Construction, 4–Way Entry
MIL–B–17757	–	Boxes, Shipping, Fiberboard (Modular Sizes)
MIL–F–21840	–	Fastener Tapes, Hook and Loop, Synthetic
MIL–T–44100	–	Thread, Para–Aramid, Spun, Intermediate Modulus
MIL–C–87261	–	Cloth, Laminated, Fire Retardant, Chemical Protective
MIL–C–87262	–	Cloth, Plain Weave, Polybenzimidazole/Aramid Blend

STANDARDS

FEDERAL

FED–STD–101	–	Test Procedures for Packaging Materials
FED–STD–595	–	Colors Used in Government Procurement
FED–STD–751	–	Stitches, Seams and Stitching

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	–	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	–	Marking for Shipment and Storage
MIL-STD-147	–	Palletized Unit Loads
MIL-STD-1668	–	Provisions For Evaluating Quality of Cloth Coveralls

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094.)

2.1.2 Other publications. The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

PUBLICATIONS

TEST METHODS

CRDC-SP-84010	–	Laboratory Methods for Evaluating the Protection of Permeable Protective Clothing Against Chemical Agents. Edgewood Arsenal

(Application for copies should be addressed to the Commander, Chemical Research and Development Center, ATTN: DRSMC-CLJ-IR (A), Aberdeen Proving Ground, Maryland 21010.)

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Test Method 135	–	Dimensional Changes in Automatic Home Laundering of Durable Press Woven or Knit Fabric

(Application for copies should be addressed to AATCC, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Copies of specifications, standards, test methods, patterns, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification contains provisions for first article inspection and approval (see 4.2 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.2 Materials

3.2.1 Overlay fabric. The overlay fabric for the CWU-66/P coveralls shall conform to MIL-C-87262 and shall be sage green USAF color Shade No. 1590 in accordance with FED-STD-595 (see 6.3).

3.2.2 Chemical agent sorptive fabric. The chemical agent sorptive fabric for the CWU-66/P coveralls shall conform to MIL-C-87261 and shall be sage green, USAF Color No. 1590 in accordance with FED-STD-595 (see 6.3). Each roll of chemical agent sorptive fabric shall be furnished with an identification and date of manufacture ticket. The garment shall be manufactured and packaged within 185 days of the ticket date.

3.2.3 Thread. The thread for all sewing operations shall be T35, 30/2 para aramid conforming to MIL-T-44100 and the color shall approximately match the color of the chemical agent sorptive fabric.

3.2.4 Elastic webbing. The elastic webbing for the waistband center section shall be class 1, 1-1/2 inches wide, minimum of 0.035 inch thick, 7 inches long, color optional of MIL-W-5664.

3.2.5 Slide fasteners. The slide fasteners shall conform to the requirements of V-F-106 and to tables I, II, and III of this specification. The chain and any other metal components except the latch and the spring of the automatic lock shall be brass with a black chemical finish. The tape of the slide fasteners shall be 3/4 inch wide, shall be high-temperature-resistant polyamide, and shall approximately match the color of the chemical agent sorptive fabric. The slide fastener for the front opening shall have a thong made of a double thickness of the overlay fabric. The hole of the pull tab of the slide fastener used on the sleeve pocket should not exceed 1/8 inch, or as an alternate, a solid pull tab may be used.

TABLE I. Slide Fasteners.

Requirements of V-F-106						Top and Bottom Tape Extensions
Location	Quantity	Type	Style	Size	Sliders 1/	(Inches)
Front opening	1	I	15	MS	2	2/
Leg openings	2	I	3	M	1	1 to 1-1/4
Breast pockets	2	I	7	M	1	1 to 1-1/4
Lower leg pockets	2	I	7	M	1	1 to 1-1/4
Left sleeve pocket	1	I	7	M	1	1 to 1-1/4

1/ The sliders shall have a short tab pull.

2/ The top tape extension shall be 1 to 1-1/4 inches. The bottom tape extension shall be 1-1/2 to 1-3/4 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE II. Length of Slide Fasteners.

Measurements in Inches 1/					
Coverall Size	Front Opening	Leg Openings	Breast Pockets	Lower Leg Pockets	Left Sleeve Pocket
32S	25	10	6–1/2	7	4–3/4
32R	26	10	6–1/2	7	4–3/4
34S	25–1/2	10	6–1/2	7	4–3/4
34R	26–1/2	10	6–1/2	7	4–3/4
36S	26	10	6–1/2	7	4–3/4
36R	27	10	6–1/2	7	4–3/4
36L	28	10	6–1/2	7	4–3/4
38S	26–1/2	10	6–1/2	7	4–3/4
38R	27–1/2	10	6–1/2	7	4–3/4
38L	28–1/2	10	6–1/2	7	4–3/4
40S	27	10	6–1/2	8	4–3/4
40R	28	10	6–1/2	8	4–3/4
40L	29	10	6–1/2	8	4–3/4
42S	27–1/2	10	6–1/2	8	4–3/4
42R	28	10	6–1/2	8	4–3/4
42L	29–1/2	10	6–1/2	8	4–3/4
44S	28	10	6–1/2	8	4–3/4
44R	29	10	6–1/2	8	4–3/4
44L	30	10	6–1/2	8	4–3/4
46S	28–1/2	10	6–1/2	8	4–3/4
46R	29–1/2	10	6–1/2	8	4–3/4
46L	30–1/2	10	6–1/2	8	4–3/4
48R	30	10	6–1/2	8	4–3/4
48L	31	10	6–1/2	8	4–3/4

1/ Except for the front opening and the breast pockets, the tolerance shall be plus or minus 3/16–inch. The tolerance for the front opening shall be plus 3/16–inch or minus 3/8–inch. The tolerance for the breast pockets shall be plus 0 or minus 3/16–inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE III. Direction of slide fastener closure.

Location	Direction of slider closure
Front opening	Top slider upward and bottom slider downward
Leg openings	Downward
Breast pockets	Downward
Lower leg pockets	Direction of outseam
Left sleeve pocket	Upward

3.2.6 Hook and pile fastener tape. The hook and pile nylon fastener tape shall conform to type I, class 1 of MIL–F–21840 and to table IV of this specification. The color of the hook and pile fastener tape shall approximately match the color of the chemical agent sorptive fabric.

3.2.7 Webbing reinforcement. The webbing used for the reinforcement piece on the pencil pockets shall conform to type XV, class 2 of MIL–W–4088. The color may be natural white. The webbing shall be cut 7 inches long.

3.3 Design. The coveralls shall be a one–piece, unlined garment with a slide fastener front closure, a bi–swing back, hook and pile fastener tape adjustments for the waist, hook and pile fastener tape closures for sleeve and leg openings, and a slide fastener size adjustment on each leg terminal. The coveralls shall have two breast patch pockets, one combination utility and multiple pencil compartment pocket on the upper front left sleeve, two lower leg patch pockets, and a multiple pencil compartment pocket on the right lower leg patch pocket. Except for the multiple pencil compartment pocket on the right lower leg, all pockets shall have butted–beaded covered slide fasteners.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IV. Fastener tapes.

Width and cut lengths of fastener tapes (inches)					
Coverall Size	Back Belt 1/ Hook	Waist 1/ Pile	Tab 2/3/ Hook	Cuff 2/3/ Pile	Nameplate Hook and Pile
32S	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
32R	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
34S	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
34R	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
36S	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
36R	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
36L	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
38S	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
38R	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
38L	1 by 4	1 by 6	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
40S	1 by 4	1 by 7	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
40R	1 by 4	1 by 7	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
40L	1 by 4	1 by 7	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
42S	1 by 4	1 by 7	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
42R	1 by 4	1 by 7	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
42L	1 by 4	1 by 7	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
44S	1 by 4	1 by 8	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
44R	1 by 4	1 by 8	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
44L	1 by 4	1 by 8	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
46S	1 by 4	1 by 8	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
46R	1 by 4	1 by 8	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
46L	1 by 4	1 by 8	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
48R	1 by 4	1 by 9	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2
48L	1 by 4	1 by 9	1–1/2 by 2	1–1/2 by 6	2 by 3–1/2

1/ Each side.

2/ Each sleeve.

3/ Each leg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.1 Component parts materials. The CWU-66/P coveralls shall be constructed from overlay fabric and chemical agent sorptive fabric. The component parts for the coverall shall be cut from the materials specified in accordance with the following pattern or standard sample parts:

Material	Pattern or Sample Parts	Cut Parts
Chemical agent sorptive fabric	Topcollar	1
	Sleeves	2
	Upper back	1
	Chest/leg front	2
	Lower back/leg back	2
	Front protective fly	1
	Front slide fastener cover	1
Overlay fabric	Undercollar	1
	Right leg pencil compartment pocket	1
	Combination utility	1
	Pencil pocket	1
	Lower leg pockets	2
	Breast pockets	2
	Sleeve and leg closure tabs	4
	Waist tabs	2
	Thong for front slide fastener	1
	Shoulder reinforcement patches	2
	Back belt	1
	Facings	2
	Leg slide fastener covers (except front slide fastener)	4
	Breast pocket zipper cover	2
	Lower leg pocket zipper cover	2
	Left sleeve pocket zipper cover	1
	Hanger	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2 Chemical agent protection. The chemical agent sorptive fabric assembly, seams and component attachment areas shall provide chemical agent protection against the agent concentration levels and exposure times given in table V when tested as specified in 4.5.1.

TABLE V. Chemical agent protection requirements.

Test	Agent	Agent Concentration (Minimum)	Exposure Time (Minimum)	Breakthrough Criteria
Vapor Agent Contamination/ Vapor Penetration	HD	20 µg/l	24 hours	4 $\mu g/cm^2$
Liquid Agent Contamination/ Vapor Permeation	HD	2 g/m^2	6 hours	4 $\mu g/cm^2$

3.4 Construction. The coveralls shall be constructed in accordance with table VI; however, the manufacturer is not required to follow the exact sequence of operations as listed therein.

3.4.1 Drill marks. Drill holes shall not be permitted.

3.4.2 Stitches, seams, and stitchings. Stitches, seams, and stitchings shall conform to FED-STD-751. Whenever two or more methods, seams, or stitches are specified for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the coveralls.

3.4.3 Fabric date tickets. Date (month and year) of manufacture of the chemical agent sorptive fabric shall appear on heavy paper tickets. The contractor shall maintain a record of fabric dates and ensure that the correct date ticket remains with each garment through cutting, sewing and packaging. Prior to packaging, the date on the ticket shall be transcribed onto the label placed in each package or printed onto the package.

3.4.4 Thread breaks and ends of all seams. Ends of seams and stitches (stitch type 301) that are not caught in other seams or stitchings shall be securely tacked or backstitched not less than 1/2-inch. When stitch type 401 is permitted as an option, the ends of stitch type 401 shall be caught in other seams or rows of stitchings. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2-inch.

3.4.5 Stitches per inch. Unless otherwise specified herein, a tolerance of plus or minus 1 stitch per inch will be permitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
1	**Cutting**			
	(a) Cut the coveralls in strict accordance with the patterns which show size, shape, placement of pockets, and notches for proper assembling of all parts. The directional line may vary from the warp direction by not more than one (1) inch on the trouser parts. The measurements shall be taken at the top and bottom edges of the pattern from the directional line to the selvage edges of the fabric; and the difference between the two measurements shall not exceed one (1) inch.			
	(b) Place the patterns on the material with the directional line coinciding with the warp threads.			
	(c) Except for the leg and sleeve tabs, the waist tabs, the front facings, the undercollar, the collar interlining, the hanger, the protective fly, and the slide fastener closure covers, which may be cut from ends, cut all parts from one piece of material.			
	(d) Replace any parts containing holes or seriously weakening defects. Holes in knit liner of chemical agent sorptive fabric can be repaired only if carbon spheres remain intact and undamaged. Holes shall be repaired using specified knit and durable, permeable, launderable adhesive. Holes through the chemical agent sorptive fabric may be repaired only in areas not visible from front or back of garment using the following procedure:			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
1	**Cutting (Continued)** i. All mends in chemical agent sorptive fabric must be made with chemical agent sorptive fabric. Damaged area shall not exceed one (1) inch. ii. Patch edges shall be folded and placed outside the area to be mended. Mend shall completely cover the damaged area with a 1/4–inch overlap. iii. Patch shall be stitched using a double row of stitching, stitch type 301, and 8 stitches per inch. iv. Mends are only permitted in areas not clearly visible (i.e., crotch, underarm, under pocket, under shoulder patch, or under back belt). A maximum of three (3) mends per garment are permitted. (e) The woven side of the chemical agent sorptive fabric is the face and shall finish on the outside of the garment.			
2	**Overedging** Overedge all slide fastener cover and pocket raw edges that are not selvaged.	502 or 503 or 504	SSa–1	8
3	**Shade marking** Except for the parts cut from ends as indicated in No. 1(c), mark all parts of the coveralls to insure a uniform shade throughout the garment. Corrosive metal–fastening devices shall not be used.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
4	**Collar**			
	(a) Join the collar to the undercollar around the outer edge with a single row of stitching 1/4–inch from the edge.	301 or 401	SSe–2(a)	12
	(b) Turn, force out the edges. Shape the rounded collar.			
5	**Sleeve, leg, and waist tabs**			
	(a) Fold the piece of overlay material on the two long sides, the pointed end of the leg/sleeve tabs and one end of the waist tabs, with a single row of stitching 1/4–inch from the edge.	301 or 401	SSe–2(a)	12
	(b) Turn, work out edges, and raise with a single row of stitching 1/4–inch from the edge.	301	SSe–2(b)	12
	Note: As an option fold the waist tab strippings in half lengthwise; turn in the raw edges, and stitch 1/16 to 1/8–inch from each edge.	301	EFp–2	12
	(c) Join the hook part of the fastener tape to the leg/sleeve tabs and the waist tabs at the pattern marks, and stitch on all four edges with a single row of stitching 1/16 inch from the edge.	301	SSa–1	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
6	**Corded beading in edges of slide fastener coverings** Fold the material to the inside at the pattern marks. Insert the cord in the fold and stitch close to the cord with a single row of stitching. The beaded edges shall be flush with each other when the slide fastener is closed. Note: All slides fastener closures for the pockets and the leg terminals, shall have corded beading in the edges of the slide fastener coverings.	301	SSbf–3(a)	12
7	**Combination utility and multiple pencil compartment pocket.** The pocket shall be a utility compartment with a covered vertical slide fastener closure, on the forward side, the full length of the pocket and of two pencil compartment openings. The compartment openings shall be 7/8–inch (plus or minus 1/8–inch) wide. The finished pocket shall be approximately 6–1/4 inches long by 3–1/2 inches wide, including the slide fastener tape. The width measurement shall be taken across the top of the pocket. Note: The top of the pencil pockets are made wider than notches on sleeve combination pocket and right leg pocket, this will give each finished pencil pocket compartment some fullness to accommodate pencils.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
7	**Combination utility and multiple pencil compartment pocket. (Continued)**			
	(a) Hem the top opening of the pencil compartment, in accordance with the notches, with a double row of stitching, with the first row of stitching 1/4–inch gauge 1/16 to 1/8–inch from the top edge catching either end of the reinforcement webbing in the stitch.	301	EFa–2	12
	(b) Fold the opening to the center notch, tacking each side edge with a single row of stitching. Stitch the webbing on the sides and bottom 1/16–inch from the edge to the pencil pocket only.	301	SSa–1	12
	(c) Join the pencil compartment to the utility pocket compartment with a double row of stitching 1/4–inch gauge 1/16 to 1/8–inch from the edge along both sides, in accordance with the notches and the pattern marks, with the bottom edge webbing longer than the pocket.	301	LSd–2	12
	(d) Place a single row of stitching centered between the front and the back edge the entire length of the pencil compartment to separate the openings. The fullness shall be distributed equally between the compartments.	301	SSv–1	12
	(e) Bartack the ends of all pencil compartment openings with the bartack superimposed on the inner row of double stitches stitching and on the center row of stitching. The bartack per per shall be 3/8 to 1/2–inch long.	Bartack		28 stitches per Bartack

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
7	**Combination utility and multiple pencil compartment pocket. (Continued)**			
	(f) Sew the slide fastener tape to the forward side of the pocket with a double row of stitching 1/8–inch to 1/4–inch gauge 1/4–inch from the edge of the scoops with the beaded edge covering the scoops.	301	LSb–2	12
	(g) Position the slide fastener cover on the opposite slide fastener tape with the beaded edge covering the scoops, and join with a single row of stitching 1/4–inch from the edge of the scoops.	301	SSa–1	12
	(h) Position the pocket to the left sleeve at the pattern marks, with the bottom raw edge of pocket turned under 3/8–inch, stitch the webbing to the sleeve on the sides and bottom 1/16–inch from edges and 1/16–inch from top of webbing folded under 1/4–inch.	301	LSd–1	12
	(i) With the sides and top raw edges turned under 3/8–inch, attach to sleeve with a single row of stitching 1/16–inch from the folded edge.	301	LSd–1	12
	(l) Place a second row of stitching 1/4–inch from the folded edge.	301	SSv–1	12
	(m) Bartack across each end of the slide fastener on the inner row of stitching.	Bartack		28 stitches per Bartack

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
8	**Fabrication of lower leg pockets**			
	(a) Sew the slide fastener tapes to the top of the pockets with a double row of stitching 1/8–inch to 1/4–inch gauge 1/4–inch from the edge of the scoops with the beaded edge covering the scoops.	301	LSb–2	12
	(b) Position the slide fastener cover on the opposite slide fastener tape with the beaded edge covering the scoops, and join with a single row of stitching 1/4–inch from the edge of the scoops.	301	SSa–1	12
9	**Right leg pencil compartment pocket**			
	(a) Hem the top opening of the pencil compartment, in accordance with the notches, with a double row of stitching, catching the reinforcement webbing in the stitch with the first row of stitching 1/16 to 1/8–inch from the top edge, 1/4–inch gauge.	301	EFa–2	12
	(b) Fold the opening to the center notch, tacking each side edge with a single row of stitching. Stitch the webbing on the sides and bottom from the edge to the pencil pocket only.	301	SSa–1	12
	(c) Position the pocket on the right leg pocket at the pattern marks, with the bottom raw edge of pocket turned under 3/8–inch, stitch the webbing to the pocket on sides and bottom 1/16–inch from edges and 1/16–inch from top of webbing folded under 1/4–inch.	301	SSv–1	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
9	**Right leg pencil compartment pocket (Continued)**			
	(d) With the sides and top raw edges turned under 3/8–inch, attach to pocket with a single row of stitching 1/16–inch from the folded edge.	301	LSd–1	12
	(e) Place a second row of stitching 1/4–inch from the sides and bottom of the pocket.	301	SSv–1	12
	(f) Place a single row of stitching, centered between the front and the back edge, the entire length of the pencil compartment to separate the openings. The fullness shall be distributed equally between the compartments.	301	SSv–1	12
	(g) Bartack the ends of all pencil compartment openings with the bartack superimposed on the inner row of double stitching on the center row of stitching.	Bartack		28 stitches per Bartack
10	**Front protective fly**			
	(a) Join the outer edges of the two plies of chemical agent sorptive fabric with a single row of stitching 3/8–inch from the edges on the top, along one side and the bottom.	301	SSe–2(a)	12
	(b) Turn the fly. Work out the edges and sew with a single row of stitching 1/4–inch from the folded edge.	301	SSe–2(b)	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
11	**Thong for front slide fastener**			
	Fabricate the thong to finish 1/2–inch wide by 3–inches long.	301	EFp–2	12
12	**Front slide fastener**			
	(a) Place the face side of the facing on the protective fly with the raw edges even. Join with a single row of stitching 3/8–inch from the edge.	301	SSa–1	12
	(b) Turn the top edge of the slide fastener cover piece under 1/4–inch. Stitch with a single row of stitching 1/16–inch from the edge, with the folded edge close to the bottom stop.	301	LSd–1	12
	(c) Position the right slide fastener on the protective fly, the front (inside) beaded edge of the coverall on the slide fastener with the raw edges even. Join, with a single row of stitching 1/4–inch from the edge of the scoops, with the front beaded edge covering the scoops.	301	LSax–4(a) (similar to)	12
	(d) Fold the facing over the stitching, and place a second row of stitching 1/16–inch from the folded edge of the facing, sewing through the coverall, the slide fastener, the protective fly, and the facing.	301	LSax–4(b) (similar to)	12
	(e) Turn the raw edge of the facing under 1/2–inch, and join with a single row of stitching 1/16–inch from the folded edge.	301	LSd–1	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
12	**Front slide fastener (Continued)**			
	(f) Position and join the left slide fastener tape the same as specified for the right slide fastener tape, except that there is no protective fly.			
	Note: The slide fastener scoops shall start 1/2–inch (plus or minus 1/4–inch) from the collar gore and extend to within 1–1/4 inches of the crotch.			
	(g) Pull the thong, through the hole in the top slider, with the raw edges even. Bartack at the bottom, the center, and up close to the slider across the 1/2–inch width.	Bartack		28 stitches per Bartack
13	**Right and left breast pockets**			
	(a) Sew the slide fastener tape to the top diagonal side of the pocket with a double row of stitching 1/16–inch from the edge, 1/8–inch to 1/4–inch gauge, 1/4–inch from the edge of the scoops.	301	LSb–2	12
	(b) Position the slide fastener cover on the opposite slide fastener tape with the beaded edge covering the scoops. Join with a single row of stitching 1/4–inch from the edge of the scoops.	301	SSa–1	12
	(c) Position the breast pockets on the right and the left and fronts at the pattern marks; turn all raw edges under 3/8–inch except the side seam, and sew with a single row of stitching 1/16–inch from the folded edge.	301	LSd–1 and SSa–1	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
13	**Right and left breast pockets (Continued)**			
	(d) Place a second row of stitching 1/4–inch from the edge at the front, top, and bottom edges. Note: A double–needle machine may be used to accomplish No. 16(c) and 16(d).	301	SSv–1	12
14	**Wearer's nameplate**			
	Position the bottom corners of the pile fastener tape nameplate above the left breast pocket, 1–inch from the front edge of the coveralls with the back bottom corner flush with the top edge of the slide fastener cover; join on all four sides with a single row of stitching 1/16–inch from the edge (see 3.4.8).	301	LSbj–1	12
15	**Upper back**			
	(a) Fold the back panel, at the inner pattern marks, face–to–face from the shoulder raw edge to the bottom edge. Join with a single row of stitching 1/16–inch from the folded edge.	301	OSf–1	12
	(b) Fold the back panel at the outer patterns marks, with the face side out, and sew with a double row of stitching 1/16–inch from the folded edge 1/4–inch to 5/16–inch gauge.	301	OSf–2	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
16	**Seat seam**			
	(a) Join with safety seam, the right panel overlapping the left panel.	515 or 516 or 519	LSq–3a	12
	(b) Turn and top stitch 1/16–inch from the overstitched edge, 1/4–inch to 5/16–inch gauge.	301	LSq–3b	8
17	**Joining of upper back to lower back**			
	(a) Join with safety seam, the upper back overlapping the front.	515 or 516 or 519	LSbm–4(a)	12
	(b) Turn and top stitch 1/16–inch from the overstitched edge, 1/4–inch to 5/16–inch gauge.	301	LSbm–4(b)	8
18	**Shoulder seams**			
	Join with double–lapped, double–stitched seams, 1/4–inch to 5/16–inch gauge, 1/16–inch from the folded edge, upper back overlapping the lower back.	301 or 401	LSc–2	12
	Note: Safety seam (516 or 519; LSbm–4(a); 12 stitches per inch) may be used. Topstitching shall be as specified in Operation 17 (b).			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
19	**Sleeves**			
	(a) Join the elbow seam of the sleeves, the topsleeve overlapping the undersleeve, with a double-lapped, double-stitched seam, 1/4-inch to 5/16-inch gauge, 1/16-inch from the folded edge.	301 or 401	LSc-2	12
	(b) Sew the sleeves to the armholes, the front and the back panels overlapping the sleeve, with a safety seam.	515 or 516 or 519	LSbm-4(a)	12
	(c) Turn and top stitch 1/16-inch from the overstitched edge, 1/4-inch to 5/16-inch gauge.	301 or 401	LSbm-4(b)	8
	(d) Serge lower raw edge of sleeve. Turn up the bottom of the sleeve at the pattern marks. Sew with a single row of stitching 1/8-inch to 1/4-inch from the edge.	502 or 503 or 504 or 301	SSa-1 EFa-1	8 12
	(e) Position the pile fastener tape at the pattern marks on the topsleeve end; sew with a single row of stitching on all four sides 1/16-inch from the edge.	301	SSa-1	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
20	**Shoulder reinforcement patch**			
	(a) Fold or form the edge of the reinforcement patch with a 3/8–inch turn–under on four sides.			
	(b) Position the curved edge of the reinforcement patch on the edge of the armhole seam, the notch matching the shoulder seam, and sew with a double row of stitching 1/16–inch from the folded edge 1/4–inch gauge.	301	LSd–2	12
	Note: As an option No. 20(c) and 20(d) may be performed in lieu of No. 20(a) and 20(b).			
	(c) Fold or form the edge of the reinforcement patch with a 1/2–inch turn–under on three sides, leaving the curved portion free. Position the curved edge of the reinforcement patch on the armhole with the center notch matching the shoulder seam. Sew the curved edge to the armhole with a single row of stitching 1/8–inch from the raw edge.	301	SSa–1	12
	(d) Stitch the folded edges of the patch to the coveralls with a double row of stitching 5/16–inch gauge 1/16–inch from the edge. The curved edge of the reinforcement patch will be caught in the armhole seam with setting in the sleeve.	301	LSd–2	12
21	**Side seams and sleeve tabs**			
	Position each tab on the sleeve raw edge at the pattern marks, pointed end toward the pile fastener tape, and stitch with a row of stitching 1/8–inch from the edge. Join the side seams as specified in No. 24, catching the tab raw edge in the side seam.	301 and 401	SSa–1	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
22	**Joining of lower leg-pocket**			
	Sew the pockets to the lower legs, in accordance with the pattern marks, with a double row of stitching 1/16-inch from the folded edge 1/4-inch gauge. Turn under 3/8-inch at the top, the bottom, and the sides.	301	LSd-2	12
23	**Leg terminal hems**			
	(a) Serge lower raw edge of legs. Turn up the bottom of the legs at the pattern marks. Sew with a single row of stitching 1/8-inch to 1/4-inch from the edge.	502 503 504 301	SSa-1 EFa-1	8 12
	(b) Position the pile fastener tape at the pattern marks on the leg terminal front with edge at the side seam. Sew with a single row of stitching on all four sides 1/16-inch from the edge.	301	SSa-1	12
24	**Inseam**			
	(a) Join with safety seam, the back overlapping the front.	515 or 516 or 519	LSbm-4(a)	12
	(b) Turn and top stitch 1/16-inch from the overstitched edge, 1/4-inch to 5/16-inch gauge.	301 or 401	LSbm-4(b)	12
25	**Leg terminal tabs**			
	Fold the raw end under 1/4-inch. Position the tab on the leg terminal at the pattern marks, the pointed end away from the inseam, with hook part of fastener tape up.	301	LSd-2	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
26	**Leg slide fastener**			
	(a) Position top of slide fastener at the bottom hem of the legs, centered over the inseam at the pattern marks with the top stops located 3/16 ± 1/16 from hem with the slide fastener tape folded under. Stitch the slide fastener with a double row of stitching 5/16–inch gage with the first row of stitching 1/8 inch from edge of tape.	301	SSv–2	12
	(b) Fold the slide fastener cover under at the top, sides and bottom and position to the outside edge of the slide fastener tape. Stitch with a double row of stitching with a 5/16–inch gage, the first row of stitching 1/8–inch from folded edge. Stitch across the bottom of slide fastener cover with a double row of stitching 5/16–inch gage with first row of stitching 1/8–inch from folded edge.	301	LSd–2	12
	(c) Catch the folded edge of the leg closure tab under both seams. The slider shall finish at the bottom of the coveralls when closed.			
	or			
	(a) Fold under top, sides and bottom of the leg slide fastener cover and position slide fastener cover to outer edge of the slide fastener tape. Stitch with a double row of stitching 1/2–inch gage with the first row of stitching 1/16–inch from the edge.	301	LSaf–2	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
26	**Leg slide fastener (Continued)**			
	(b) Position slide fastener cover and slide fastener to bottom of leg centered over the inseam at the pattern marks and stitch top, sides and bottom with a double row of stitching 5/16–inch gage with the first row of stitching 1/8–inch from edge.	301	SSv–2	12
	(c) Catch the folded edge of the leg closure tab under the seam. The slide fastener shall be at the bottom of the coverall when closed.			
27	**Belt assembly**			
	(a) Fold the raw end of the waist adjustment tabs under 1/2–inch. Position on the elastic webbing by overlapping 1/2–inch on each end, and join with a single row of stitching, forming a box and "X" type stitch 3/8–inch by 7/8–inch.	301	LSd–1	12
	(b) Center the elastic webbing on the center of the back waist seam. Place a vertical bartack through the elastic webbing stitches and the coveralls at the waist and seat seam.	Bartack		28 stitches per Bartack
	(c) Double fold the ends of the back belt to the inside 1/4–inch, and sew with a single row of stitching 1/16–inch from the folded edge.	301	EFb–1	12
	(d) Turn under each raw edge of the back belt lengthwise, and sew to the back of the coveralls, centered over the waist seam and the pattern mark at the center back with a double row of stitching 1/16–inch from the folded edges 1/4–inch gauge.	301	LSd–2	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
27	**Belt assembly (Continued)**			
	(e) Position the waist pile fastener tapes on the front panels at the waistline pattern marks, centered over the lower folded edge of the breast pockets, with the front edge even with the breast pocket front edge. Sew with a single row of stitching 1/16–inch from the edge on four sides. The pile fastener tapes shall be aligned with the waist tabs.	301	LSbj–1	12
28	**Hanger**			
	(a) Fabricate the hanger to finish 1/2–inch wide by 4–inches long.	301	EFp–2	12
	(b) Position the hanger at the center of the back of the collar. Hanger ends should be spaced 2 inches apart and caught in collar seam forming a loop.			
29	**Joining of collar to neck**			
	(a) Sew the undercollar to the neck with a single row of stitching 3/8–inch from the edges.	301	SSa–1	12
	(b) Turn under raw edges of topcollar and sew to the neck with a single row of stitching 1/16–inch from the folded edges catching both hanger ends.	301	LSd–1	12
	(c) Raise stitch the collar with a single row of stitching 1/4–inch from the outer edge all around the collar.	301	SSe–2(b)	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
30	**Label**			
	Position the contractor/instruction label approximately 1–1/2 inches from the neck seam, centered directly below the hanger, on the inside back of the coveralls. Sew the label securely along all four (4) edges to the coveralls with a single row of stitching 1/16–inch from the edge. Position the laundry instruction label, centered directly below contractor/instruction label, and sew securely along all four (4) edges to the coveralls with a single row of stitching 1/16–inch from the edge.	301	SSa–1	12
31	**Bartack**			
	The bartacks shall be 3/8–inch to 1/2–inch in length and positioned as follows: (1) One at the crotch seam, catching the slide fastener tapes. (2) One across both slide fastener tapes, centered between the crotch seam and the bottom of the front opening slide fastener bottom stop, uniting the edges of both tapes catching the protective fly. (3) One across the bottom stitching on each side of the leg slide fastener tape at the leg terminals and one across the top closed end on the inside row.	Bartack		28 stitches per Bartack

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Sewing operations – Continued.

No.	Description of operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
31	**Bartack (Continued)** (4) Two on each breast pocket: one at each end of the slide fastener on the inside row of stitching. (5) Two on each lower leg pocket, one at each end of the slide fastener on the inside row of stitching. (6) Two on each end of the back belt on the inside row of stitching. (7) One on each shoulder reinforcement patch at the top of the bi–swing back pleats on the outside row of stitching. (8) One across the double row of stitching on the trouser leg inseams, 1/8–inch from the leg terminals. (9) One across the double row of stitching on the bottom edge of the sleeve side seam, 1/8–inch from the bottom edge.			
32	**Cleaning** Trim all thread ends, and remove all loose thread ends. Remove all spots, stains, and shade tickets without injury to the fabric.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.4.6 Sewing tolerances. The sewing tolerances for 1/4–inch and 5/16–inch gauges of stitching shall be plus or minus 1/16–inch. The sewing tolerances for 3/8–inch gauge of stitching shall be plus 0 or minus 1/8–inch. The sewing tolerance for 1/16–inch from the edge shall be minus 0 or plus 1/32–inch.

NOTE: Top stitching option of 3/16–inch to 1/4–inch gauge may be used if consistent throughout garment

3.4.7 Corded beading in edges of slide fastener coverings. All slide fastener closures for the pockets, leg terminals, and front opening shall have a cord beading in the edges of the slide fastener coverings conforming to Type IA of MIL–C–5040. The outer beaded cloth edges of all slide fastener closures shall be flush with the slide fastener scoops. The scoops, when locked together, shall be completely concealed by the beaded coverings.

3.4.8 Hook part of fastener tape for nameplate. An identical size of the hook fastener tape shall be affixed to the pile fastener tape for use as a mounting for the wearer's nameplate (see No. 14 of table VI).

3.5 Patterns. The manufacturer's working patterns shall be identical in size and shape to the patterns referenced by this specification. Unless otherwise specified herein, the patterns provide a allowance of 3/8–inch for single–needle seams and 1/2–inch for double–needle seams.

3.6 Finished measurements. The finished measurements of the coveralls shall conform to table VII.

3.7 Labels

3.7.1 Contractor/general instruction label

a. Each of the coveralls shall have a combination size/identification/instruction label conforming to type I, class 14 of DDD–L–20, except that the name of the contractor and the specification number shall be included on the label. The label shall have colorfastness to laundering.

b. The instruction label shall conform to the requirement of type I, class 14 of DDD–L–20. The body of the instruction label shall be as follows:

COVERALLS, CHEMICAL DEFENSE CWU–66/P

1. Wear as outer garment.

2. WARNING. IMPORTANT. USE MINIMUM AMOUNT OF DETERGENT RECORD NUMBER OF LAUNDERINGS. MACHINE LAUNDER DELICATE CYCLE ONLY, WARM WATER. TUMBLE DRY, PERMANENT PRESS OR AIR SETTING. DO NOT DRY CLEAN.

3. Launder with slide fasteners and hook and pile fasteners closed. Do not press garment.

4. Lubricate slide fasteners with Bees Wax.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VII. Finished measurements of coveralls.

Finished Measurements in Inches 1/ 2/			
Coverall' Size	Breast	Sleeve Inseam Length	Leg Inseam Length
32S	17–3/4	20–1/4	28–1/2
32R	17–3/4	22	30–1/2
34S	18–1/2	20–1/4	28–1/2
34R	18–1/2	22	30–1/2
36S	19–3/4	20–3/4	28–1/2
36R	19–3/4	22	30–1/2
36L	19–3/4	23	32–1/2
38S	21	20–3/4	28–1/2
38R	21	22	30–1/2
38L	21	23	32–1/2
40S	22–1/2	21	28–1/2
40R	22–1/2	22	30–1/2
40L	22–1/2	23	32–1/2
42S	23–1/2	21–1/4	28–1/2
42R	23–1/2	22–1/4	30–1/2
42L	23–1/2	23–1/4	32–1/2
44S	24–1/2	21–1/4	28–1/2
44R	24–1/2	22–1/4	30–1/2
44L	24–1/2	23–1/4	32–1/2
46S	25–1/2	21–1/4	28–1/2
46R	25–1/2	22–1/4	30–1/2
46L	25–1/2	23–1/4	32–1/2
48R	26–1/2	22–1/4	30–1/2
48L	26–1/2	23–1/4	32–1/2

1 A tolerance of plus or minus 1/2–inch will be permitted.

2/ The procedure for measuring shall be as follows: The coveralls to be measured shall be closed; the hook and the pile parts of the fastener tapes shall not be engaged. The coveralls shall be laid flat without tension on a smooth flat surface so that the creases and wrinkles shall not affect the measurement. The measurement of the breast shall be taken across the breast area, from folded edge to folded edge, at the bottom of the armholes. The measurement of the sleeve inseam length shall be taken, along the inseam on the sleeve, from the base of the armhole to the bottom of the sleeve. The measurement of the leg inseam length shall be taken, along the inseam on the leg, from the center of the crotch seam to the bottom of the leg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

5. For thorough cleaning, return unit for machine washing in accordance with established laundry procedures. (DELICATE CYCLE, WARM WATER). Coveralls may be hand laundered using mild soap and warm water. All soap should be thoroughly rinsed out.

6. Do not starch or bleach.

DO NOT REMOVE THIS LABEL

c. The label shall be positioned as specified in table VI operation 30.

3.7.2 Laundry Instruction Label. A laundry instruction label shall be centered below the contractor/general instruction label and stitched on all four (4) sides. The label shall have colorfastness to laundering. The laundry instruction label shall read:

LAUNDERING INSTRUCTIONS
LAUNDER WITH MILD SOAP IN LUKE WARM WATER.
THE ENSEMBLE MAY BE WASHED UP TO TEN TIMES.
LINE THROUGH THE NEXT UNUSED NUMBER.
DO NOT DRY CLEAN.

01	02	03	04	05	06	07	08	09	10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation, is an acceptable practice to ascertain compliance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document (see 2.1, 2.2).

4.4 First article inspection. The first article inspection shall consist of specified examinations and tests performed to determine that the production item conforms to the requirements of this specification.

4.4.1 First article samples. Unless otherwise specified, as soon as practical after the award of the contract or order, the manufacturer shall submit one hundred (100) completed garments for subjection to the examinations and tests specified in 4.4.2. The samples shall be representative of the construction, workmanship, components, and materials to be used during production.

4.4.2 First article tests. The first article sample shall be subjected to tests specified in 4.5.

4.4.2.1 Testing of first article suits. For first article testing, ten (10) suits shall be submitted for all chemical agent protection testing in accordance with table V. Testing shall be performed as specified in 4.5.1.

4.5 Quality conformance tests. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL–STD–105.

4.5.1 Chemical agent protection. Compliance with chemical agent protection requirements of table V shall be determined on suits as shown in table VIII. Suits shall be submitted only to government approved laboratories (see 6.5) for testing. Suits shall be tested after ten (10) AATCC Test Method 135–1987 launderings and dryings using delicate wash cycle, 41 degrees ±3° C water temperature, 15 grams AATCC standard detergent #124 with optical brightener and permanent press tumble dry procedure. Garments shall be tested in any area of chemical agent sorptive fabric including seams and/or attachments. The sample unit shall consist of one suit. Two samples of 3 specimens each shall be cut from each garment. One sample shall be tested for HD vapor contamination/vapor penetration and one sample shall be tested for HD liquid agent contamination/vapor permeation. Any failure shall be cause for rejection of the suits, represented by the sample size. The sample size shall be in accordance with MIL–STD–105.

4.5.2 Components and materials. Unless otherwise specified herein or in the applicable purchase documents (see 6.2), components and materials shall be tested in accordance with all the requirements of referenced documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE VIII. Chemical agent protection test methods.

Test	Agent	Test Method	Reference
Vapor Agent Contamination/ Vapor Penetration	HD	2.1	CRDC–SP–84010
Liquid Agent Contamination/ Vapor Permeation	HD	2.2	CRDC–SP–84010

4.5.3 End item inspection. Each end item shall be visually examined as specified in 4.5.4, 4.5.5, 4.5.6, and 4.5.7. The sample unit for these examinations shall be one coverall.

4.5.4 Methods of inspection.

4.5.4.1 Critical defect examination. Every suit in each lot shall be examined for critical defects listed in table IX. Any suit found to contain one (1) or more critical defects shall be rejected.

4.5.4.2 Visual examination. The end item shall be visually examined for defects listed in table IX. Lot size shall be expressed in units of suits. For all defects, the inspection level shall be II, and the acceptable quality level (AQL) shall be 4.0 major and 15 total defects per hundred units, in accordance with MIL–STD–105. Any critical defect found during this examination shall be cause for the immediate rejection of the lot.

4.5.5 End item dimensional examination. The end item shall be examined for dimensional defects as specified in MIL–STD–1668 unless otherwise specified.

4.5.6 Examination of packaging requirements. The packaged items shall be examined as specified in MIL–STD–1668 unless otherwise specified.

4.5.6.1 Package examination. Each filled and sealed package shall be examined for defects listed below. The occurrence of any defect shall be cause for rejection of the package. Garments from rejected packages may be repackaged as specified in 5.1 and submitted for re–examination.

Defect

Closure seal width not as specified
Closure seal not continuous
Tear, cut or hole in package

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects.

Defect	Classification		
	Critical	Major	Minor
Material Defects and Damages			
1. Abrasion damaging the carbon layer of the chemical agent sorptive fabric, non-coated area, or area of embedded foreign matter on the chemical agent sorptive fabric	X		
2. Any cut, hole, tear, burn through, or improper mend on chemical agent sorptive fabric	X		
3. Any defect affecting appearance, but not serviceability, such as excessive shading, thick place, or color differing from that of standard sample			X
4. Soiling and staining of fabric by oil or other contaminating substance			
a) Chemical agent sorptive fabric	X		
b) Overlay fabric			X
5. Cleanness – thread ends not trimmed or loose thread ends not removed when condition exists over major portion of coverall.			X
6. Material outdated (more than 185 days past ticket date on unpackaged garments)	X		
7. Material deviating from specification requirements			
a) Chemical agent sorptive fabric	X		
b) Overlay fabric			X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Components and Assembly			
1. Any design, component part, or operation omitted or deviating from specification requirements (unless otherwise classified herein)		X	
2. Improper garment assembly utilizing overlay fabric for any of the following components: topcollar sleeves upper back chest/leg front lower/leg back front protective fly front slide fastener cover	X		
3. Any component part tight, twisted, distorted, puckered, or pleated		X	
4. Wrong component used in assembly (i.e., sleeves reversed)		X	
Cutting			
Any component part not cut in accordance with specified pattern, directional lines on patterns; or not in accordance with specification requirements	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Seams and Stitching			
1. Accuracy of seaming			
a) Seam twisted or pleated		X	
b) Part of garment caught in any unrelated operation or stitching		X	
c) Ends of seam not secured in other stitching or not backstitched at least 1/4–inch		X	
d) Any line of stitching omitted or misplaced (not otherwise classified)		X	
2. Gauge of seams or seam allowance not within specified range or tolerances, on major portion of seam		X	
3. Stitch type or seam type other than specified		X	
4. Open seam (2 or more skipped broken, runoff, or missing stitches)			
a) On joining seam of chemical agent sorptive fabric	X		
b) On other seams, including pocket assembly, patches			
– Less than 1/2–inch			X
– 1/2–inch or more		X	
5. Runoffs			
a) On joining seams, except when resulting in open seam (use "open seam" defect)			
b) Edge or top stitching more than 1/4–inch (when not resulting in open seam)			X
6. Raw Edges			
a) On outside			
– Less than 1/2–inch			X
– 1/2–inch or more		X	
b) On inside and hems			
– Less than 1–inch			X
– 1–inch or more		X	
Note: Raw edge not securely caught in stitching shall be classified as an open seam.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Seams and Stitching (Continued)			
7. Stitch tension			
a) Loose tension for up to 1–inch			X
b) Loose tension for more than 1–inch		X	
c) Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X		
8. Stitches per inch (score only when condition exists on more than 1/2 the length of the seam)			
a) Less than specified			
– By one or two stitches			X
– By more than two stitches		X	
b) One or more stitches more than specified, not damaging cloth			X
c) More than specified, damaging woven fabric or carbon spheres layer of chemical agent of sorptive fabric	X		
9. Stitches skipped or broken, not resulting in open seam			
a) Up to a length of 1/4–inch			X
b) 1/4–inch or more		X	
10. Bartack			
a) Missing, insecure, misplaced, not serving intended purpose, or not specified size or type:			
(1) For attachment		X	
(2) For reinforcement			X
b) Loose stitch tension			X
c) Length not as specified by 1/16–inch or more			X
d) Direction not as specified (vertical or horizontal)			X
e) Number of stitches less than specified by three (3) or more			X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Fastener Tapes (hook and pile)			
1. Omitted	X		
2. Misplaced or not positioned as specified, i.e., not serving intended purpose	X		
3. Type, color, or width not as specified			X
4. Stitched too loose or too tight causing noticeable puckering or twist on outside			X
Slide Fasteners			
1. Any part of metal fastener assembly defective or missing, scoops of chain unevenly matched or stitching too close to metal edge, not permitting slider to pass freely	X		
2. Cut or tear in tape		X	
3. Not positioned as specified		X	
4. Tape set too loose or too tight causing noticeable bulge, or puckering on closed front or pocket	X		
5. Not functioning properly	X		
6. Tape not attached as specified		X	
7. Ends of chain at top uneven by 3/16-inch or more			X
8. Thong omitted on top slider		X	
9. Any untrimmed or loose threads that impair the ease of operations of any slide fastener		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Reinforcement Pieces			
1. Omitted	X		
2. Misplaced, not serving intended purpose			X
3. Length or size not as specified			X
Labels			
1. Omitted		X	
2. Inscription incomplete, incorrect or illegible			X
3. Insecurely attached or not stitched on all four sides			X
4. Position not as specified			X
Hanger Loop			
1. Omitted		X	
2. Spread measurement between ends not as specified			X
3. Misplaced or not attached as specified			X
4. Length or width not as specified			X
5. Not securely caught in seam			X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Tabs (leg, waist, sleeve)			
1. Too tight, loose or twisted when fastened			X
2. Irregular in width by 3/16–inch or more			X
3. Finished length not as specified			X
4. Misplaced or not positioned as specified		X	
5. Constructed in other than specified manner		X	
6. Insecurely caught in stitching		X	
7. Omitted	X		
8. Unequal distance from seam or edge			X
Sleeves			
1. Sleeves reversed	X		
2. Pleated or excessive fullness at armhole		X	
3. Tight at armhole causing puckers on front or back			X
4. Width of cuff not as specified			X
5. Cuff, twisted or puckered			X
Hems (sleeve, leg pocket)			
1. Finished width not as specified			X
2. Irregular in width by 1/4–inch or more on more than 1/2 the length of seam			X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Classification of defects – Continued.

Defect	Classification		
	Critical	Major	Minor
Collar			
1. Collar off center 1/2–inch or more than shoulder seams at neckline are matched		X	
2. Edge not forced out			X
Front Closure			
1. Front noticeably twisted, puckered, or bulging when fastened		X	
2. Uneven at neck by 1/2–inch or more when closed		X	
3. Misplaced protective fly	X		
4. Protective fly not secured at lower edge by bartack			X
Pockets and Flaps			
1. Omitted	X		
2. Finished length or width not as specified			X
3. Not positioned as specified a) By 1/2–inch or less b) By more than 1/2–inch		 X	 X
4. Corner of pocket not securely tacked			X
5. Noticeably twisted or puckered			X
6. Construction not as specified			X
Templates			
Any pattern part varying by a total of 1/4–inch but not more than 1/8–inch on any one side of the template		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.5.6.2 Package leakage testing. Each package prior to over bagging and packing shall be left to stand a minimum of 12 hours after sealing. Following the 12-hour period each package shall be visually inspected for leakage. Any package that exhibits an increase in volume or soft hand where the vapor barrier bag is easily separated from the garment shall be considered a failure. Any failure shall be repacked and resubmitted for visual retesting. Daily, a minimum of 20 packages that pass the visual leakage test shall be tested in accordance with the hot water technique of method 5009.3 of FED-STD-101. Bubbles which may appear on the package surface but are not released, or are released at a decreasing rate shall not be construed as an indication of failure. A steady stream or recurring succession of bubbles from any surface or seam shall be considered a failure. Any failure shall be cause for leak testing to the hot water technique of each package in the lot and disposal of the failed packages. All packages shall be thoroughly dried prior to being sealed in a polyethylene pouch (see 5.1.3).

4.5.6.3 In-process test of heat sealed closure of seam. The test shall be made daily prior to the start of filling and sealing packages to determine that the heat sealing equipment is producing seams in accordance with the requirements of 5.1.2. One full seam shall be sealed on an empty barrier bag using recommended heat seal settings and tested in accordance with method 2024 of FED-STD-101 at normal room temperature with a static load of 50 ounces using a fabricated bag specimen procedure. A 5 percent reduction in static load is permitted when room temperature in the test area exceeds 90° F. Any test failure shall be cause for readjustment of the sealer settings and retesting until a satisfactory seam is produced.

4.5.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one (1) palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the acceptable quality (AQL), expressed in defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified
	Interlocking of loads not as specified
	Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Folding. Each of the coveralls shall have all fastenings secured and shall be neatly folded to an approximate size of 12 inches in length by 15 inches in width. All slide fasteners shall be located on the inside of the folded garment to prohibit contact with barrier bag. Garments shall be held in their folded position with a rubber band at each end. Rubber bands shall be of suitable length and width for the purpose intended.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

5.1.2 Sealing. The folded coveralls shall be individually placed in a heat sealable, flexible, water vapor proof, greaseproof barrier bag conforming to type I, class E, style 1 of MIL–B–117 using MIL–B–131, class 1 tyvek exterior materials only. The bags shall have a nick tear, or V–notch on side edge as specified in MIL–B–117. The size of each bag shall be such that the garments fit snugly therein (approximately 17 inches by 22 inches). The bag shall be flat style. Each bag shall have all air evacuated and be heat sealed while maintaining a vacuum of 26 to 28 inches of mercury. The closed package shall displace between 200 and 400 cubic inches of water when tested as specified in 4.5.6.2 without damaging the garment. All packages shall be thoroughly dried prior to being sealed in the polyethylene pouch. The seal shall be 1/2 (±1/16) inch wide when performed on a jaw type sealer. Closure effected by the dielectric, impulse or ultrasonic process shall be 3/16 (± 1/16) inch wide. The sealed package shall not leak when tested in accordance with 4.5.6.3. Packages given the hot water leakage test shall be thoroughly dried prior to being sealed in the polyethylene pouch. Garments removed from their packages in accordance with MIL–STD–1668 shall show no evidence of damage when inspected visually in accordance with 4.5.3. A garment, if non–defective, shall be repackaged and retested in accordance with 5.1.2 and 4.5.6.2.

5.1.3 Bagging. Every vacuum sealed barrier bag shall be inserted into a clear polyethylene pouch of equivalent size, and heat sealed. The polyethylene pouch shall be fabricated from tubing or sheet stock with a minimum thickness of 0.004 inches and be provided with a 1/4–inch diameter hole near the bottom for excess air to escape.

5.1.3.1 User procedures. Each polyethylene barrier bag shall contain one (1) copy of user procedures (see 6.6). User procedure shall be placed so that unit package markings and user procedure can be read without opening barrier bag.

5.2 Packing. Packing shall be level A and B as specified (see 6.2).

5.2.1 Level A. Ten (10) coveralls of one size and color only, packaged as specified, shall be packed in a fiberboard shipping container assembled, closed, and reinforced to conform to class weather–resistant, grade W5c, size 3A of MIL–B–17757. Approximate inside dimensions of each shipping container shall be 23–1/2–inches in length, 15 inches in depth and 15 inches in height. Packages shall be packed on end. All voids shall be filled with commercial grade kraft paper.

5.2.2 Level B. Ten (10) coveralls of one size and color only, packaged as specified, shall be packed in a fiberboard shipping container assembled and closed in accordance with MIL–B–17757. The container shall be size 3A constructed of grade 275 fiberboard of MIL–B–17757. Approximate inside dimensions of each shipping container shall be 23–1/2 inches in length, 15–inches in depth, and 15–inches in height. Packages shall be packed on end. All voids shall be filled with commercial grade kraft paper.

5.2.3 Palletization. When specified, (see 6.2), suits packed as specified in 5.2 shall be palletized on a 4–way entry pallet in accordance with load type Ia of MIL–STD–147. Each prepared load shall be bonded with horizontal, primary, and secondary straps in accordance with the bonding method C and D, or bonding methods F or G. Pallet pattern shall be in accordance with the appendix of MIL–STD–147. The pallet shall be 4–way, type I, class 1, style 1, size A, wood group I, II, III, or IV of MIL–P–15011, or 4–way type IV, V, or VIII, class 1, style A, size 2, wood group I, II, III, or IV, grade A of NN–P–71. Each pallet load shall contain the same number of cartons.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special marking required by contract or order, unit packages and shipping containers shall be marked in accordance with MIL–STD–129 and shall include the size of the coveralls, the garment production date (month and year), fabric production date (month and year), and type II shelf life. The shelf life shall be 10 years from the garment production date. The expiration date shall be expressed by month and year.

6. NOTES

6.1 Intended use. The coveralls covered by this specification are intended to be worn as an outer garment by flight personnel for protection from chemical warfare agents.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article (see 3.1 and 4.2).

c. Type and size required (see 1.2.1 and 1.2.2).

d. Requirements for testing components and materials (see 4.2).

e. Selection of applicable levels of packing (see 5.2).

6.3 Color sample. Sample of color shade may be obtained from the procuring activity or as directed by the contracting officer.

6.4 Figures. Figures 1 and 2 show the general style of the coveralls and are furnished for information only.

6.5 Chemical agent protection. A list of government approved laboratories for chemical agent testing is available from Commander, Chemical Research and Development Center, ATTN: SMCCR–CMC, Aberdeen Proving Ground, Maryland 21010.

6.6 User Procedures. Copy of user procedure maybe obtained from the procuring activity or as directed by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

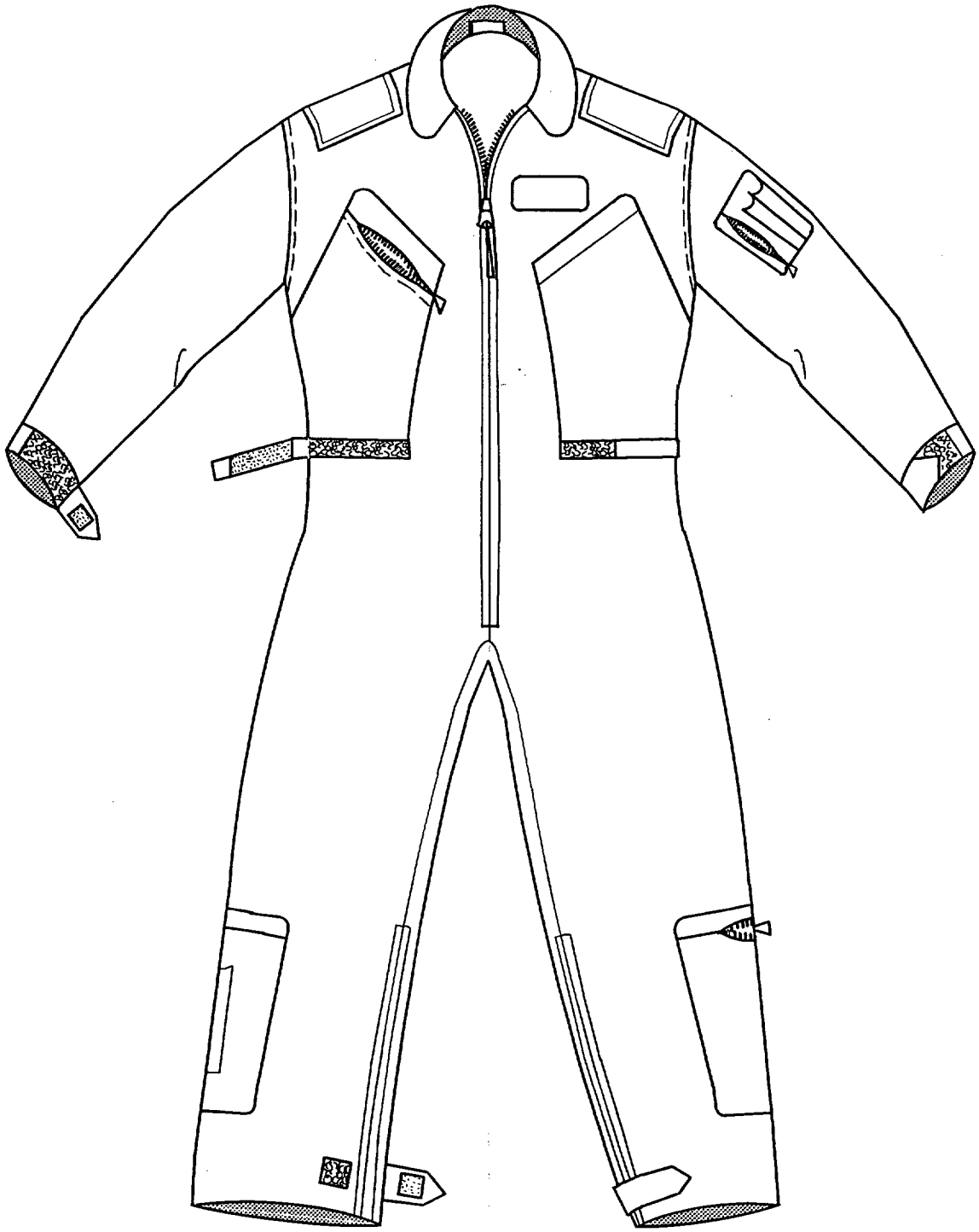


FIGURE 1. General style of coverall — front view.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

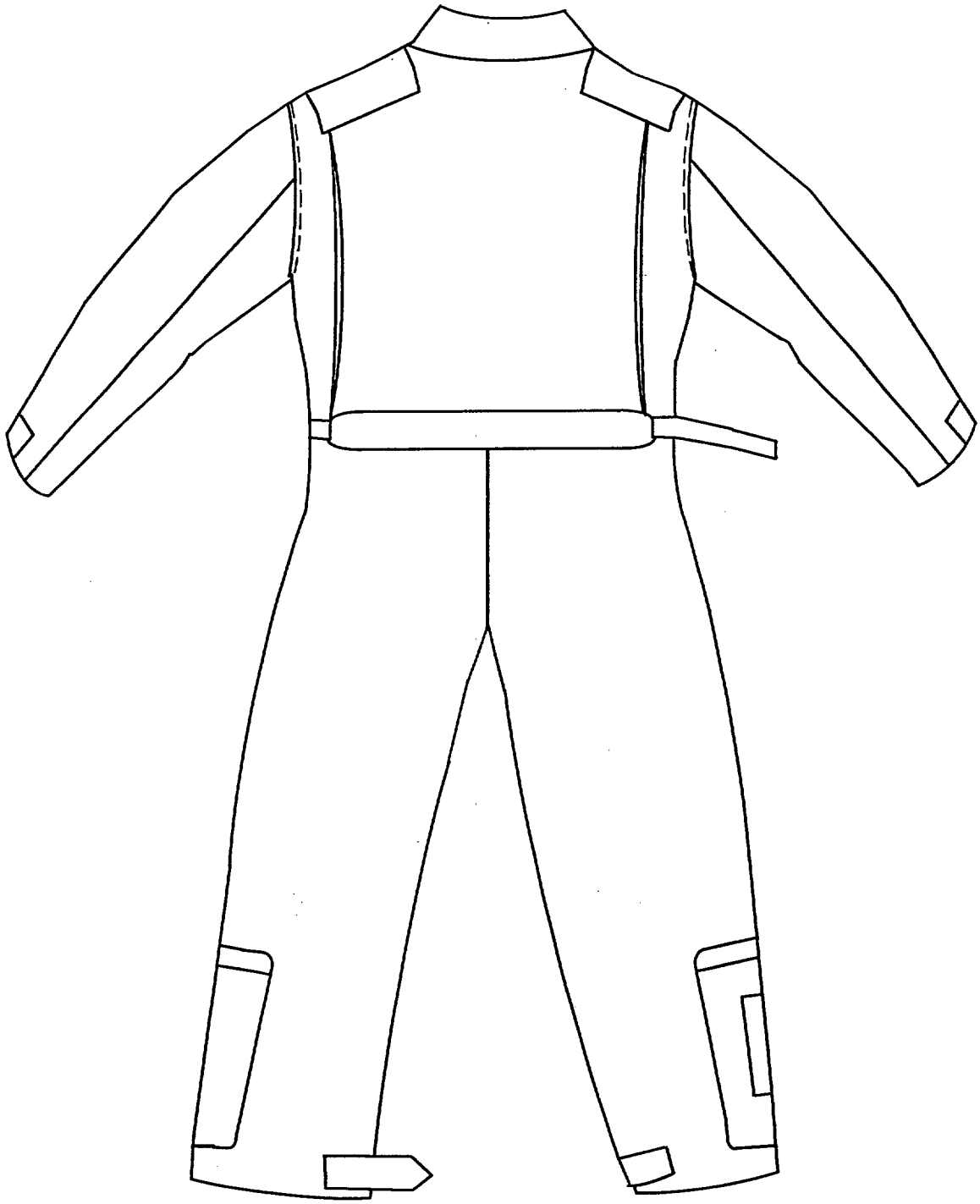


FIGURE 2. General style of coverall — back view.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

6.7 Subject items (key word) listing.

Aircrew
Coveralls, chemical protective
Carbon activated
Chemical agent sorptive fabric
Cloth, Polybenzimidazole/Aramid
Elastic webbing
Fire–resistant
Nylon fastener tape
Overlay fabric
Slide fasteners
Thread
Webbing

Custodian:
Air Force – 11

Review Activity:
Air Force – 82
DLA–CT

Preparing Activity:
Air Force – 11

Project No. 8415–F853

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, not to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractural requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-87264 (USAF)	2. DOCUMENT DATE *(YYMMDD)* 920211

3. DOCUMENT TITLE
Coveralls, Chemical Defense CWU-66/P

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

A. NAME AF CODE 11	B. TELEPHONE *(Include Area Code)* (1) Commercial: (513) 255-6281 (2) AUTOVON *(If applicable)*: DSN 785-6281
C. ADDRESS *(Include Zip Code)* ASD/ENES WRIGHT-PATTERSON AFB OH 45433-6503	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.